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INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A REPORT ON
NEPAL TRAFFIC ACCIDENT PATTERN ANALYSIS

SUBMITTED BY:

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We would also like to convey our gratitude to our colleagues and supervisors for their support and encouragement. This project was more than just an academic task; it was an experience that enabled us to apply our practical skills in Data Science, particularly in spatial analysis, geographic clustering, and data visualization using Python. Working with DBSCAN, Kernel Density Estimation, and Moran's I spatial autocorrelation gave us valuable hands-on exposure to real-world problem solving in the context of road safety in Nepal. The educational experience and practical application have been incredibly rewarding, and we appreciate everyone who contributed to this learning journey.

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ABSTRACT

Road traffic accidents are a growing public safety challenge in Nepal, with fatalities rising steadily across all seven provinces. This study presents a spatio-temporal analysis of 5,000 road accident records spanning 2021 to 2025, with the objective of identifying geographic hotspots, uncovering temporal patterns, and providing evidence-based recommendations for road safety interventions.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) with the Haversine distance metric was applied to identify 9 distinct accident clusters across Nepal's major districts and highway corridors. Kernel Density Estimation (KDE) was used to generate continuous accident density surfaces, and Moran's I spatial autocorrelation was computed to statistically validate the clustering of accident severity and risk scores.

The results identify distinct provincial-level differences in fatality risk, with behavioral causes — particularly human error, overspeeding, and drunk driving — dominating accident patterns nationally. Motorcycles are the most involved vehicle type, and night-time is consistently the highest-risk period. Spatial autocorrelation testing confirmed that high-risk accidents are geographically concentrated rather than randomly distributed, validating the hotspot findings.

The findings provide location-specific insights to guide targeted road safety investments, enforcement priorities, and emergency response improvements across Nepal's provinces.

Keywords: *Road Traffic Accidents, DBSCAN, Spatial Hotspot Analysis, Nepal, Spatio-temporal Analysis, Kernel Density Estimation, Moran's I, Motorcycle Safety, Cross-Validation*

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1. INTRODUCTION AND PROBLEM STATEMENT

1.1 Background

Road traffic accidents are a global public health crisis. The World Health Organization (WHO) estimates approximately 1.19 million deaths annually from road traffic crashes, with 20–50 million sustaining non-fatal injuries [1]. The burden falls disproportionately on low- and middle-income countries, where fatality rates are three times higher than in high-income nations.

Nepal, as a rapidly developing country with exponentially growing vehicle registration, faces an acute road safety challenge. The country's diverse geography — ranging from flat Terai plains at 60m elevation to high Himalayan terrain exceeding 8,000m — combined with monsoon-season hazards such as landslides and floods, creates a uniquely complex accident environment. Nepal's road infrastructure, emergency response systems, and traffic enforcement mechanisms are still developing, further compounding risk.

1.2 Problem Statement

Road traffic accidents remain a significant public safety issue in Nepal, with incidents occurring across different regions, road types, and environmental conditions. However, accident data is often analyzed only through simple statistics, making it difficult to identify geographic patterns and high-risk locations. Without proper spatial analysis, policymakers and traffic authorities cannot effectively prioritize areas for safety interventions.

This project aims to analyze road accident data using spatial data analysis techniques to identify accident hotspots and examine how factors such as road type, weather conditions, and time of day influence accident occurrence. By applying clustering and visualization methods, the study seeks to uncover meaningful patterns that can support data-driven road safety planning.

1.3 Objectives

- To analyze road traffic accident data in Nepal using data science techniques to identify spatial patterns and high-risk accident locations.

•To detect accident hotspots using spatial clustering methods such as **DBSCAN** and analyze how factors like **time of day, road type, and weather conditions** influence accident occurrence.

1.4 Scope of Study

- 5,000 road traffic accident records from 2021 to 2025 across all 7 provinces of Nepal.
- The dataset includes variables such as latitude and longitude, time of day, road type, weather condition, vehicle type, and accident cause.
- Spatial clustering using DBSCAN with Haversine metric for GPS coordinate data.
- Pattern analysis of causes, vehicles, weather conditions, time of day, and demographics.
- Comparative severity and emergency response analysis across provinces.

1.5 Significance of Study

- Providing the first province-wide spatial clustering analysis of Nepal road accidents using DBSCAN.
- Generating actionable, location-specific insights that can guide infrastructure investment, enforcement, and emergency service deployment.
- Demonstrating the applicability of modern data science methods (Python, scikit-learn, Folium) to Nepal's road safety context.

1.6 Limitations

- The dataset is **synthetic**, so it may not perfectly represent real-world accident patterns in Nepal.
- Accident data is derived from a structured dataset and may not capture all unreported incidents.
- DBSCAN clustering is based on geographic proximity (GPS coordinates) and may group accidents from adjacent sub-districts.
- Causal relationships (e.g., road condition causing an accident) are based on reported data and cannot be independently verified.

2. LITERATURE REVIEW

2.1 Spatial Analysis Methods for Accident Hotspot Detection

Spatial analysis of road traffic accidents has become an established approach for identifying crash concentration zones and guiding safety interventions. Kernel Density Estimation (KDE) is widely used to generate continuous accident density surfaces, while DBSCAN (Density-Based Spatial Clustering of Applications with Noise) has gained adoption for identifying arbitrarily shaped hotspot clusters without requiring a pre-defined number of clusters. Unlike K-Means, DBSCAN classifies outliers as noise — particularly useful in geographic accident analysis where isolated incidents should not be forced into clusters. Moran's I spatial autocorrelation complements these methods by statistically testing whether accident severity values cluster geographically or are randomly distributed [2, 3].

2.2 Studies in the Nepali Context

Within Nepal, several studies have applied spatial methods to road accident data. Aryal and Dhakal (2024) visualized accident data in Bhaktapur district using ArcGIS, identifying Thimi, Radhe Radhe, and Sallaghari as persistent hotspots on the Araniko Highway and finding motorcycles and overspeeding as dominant factors [4]. Bhele et al. (2024) conducted a GIS-based spatial and temporal analysis in Banepa, Panauti, and Dhulikhel municipalities, applying KDE hotspot mapping and temporal analysis to confirm accident concentration along the Araniko Highway corridor, with major accidents predominantly in Banepa [5]. Anuradha K.C. (2024) conducted the most detailed valley-level study using Moran's I, Getis-Ord Gi*, KDE, and COMAP techniques in Kathmandu Valley, finding morning and evening peak periods as highest risk [6]. Mahato et al. (2025) extended spatial analysis to all 77 districts of Nepal using LISA and Global Moran's I, confirming persistent High-High clusters in Kathmandu, Lalitpur, and Bhaktapur, with motorcycles as the dominant vehicle type in accident clusters [7]. Bhatta et al. (2025) applied KDE and Global Moran's I to the Prithvi Highway (2019–2025), identifying persistent hotspots at Kalanki–Thankot, Kurintar–Mugling, and Aabukhaireni–Damauli, and finding that crash frequency and injuries show statistically significant spatial clustering while fatalities exhibit a random pattern influenced by situational rather than locational factors [8].

2.3 Research Gap and Contribution

Existing Nepal-specific studies are either confined to the Kathmandu Valley or focus on individual highway corridors. None apply DBSCAN clustering with the Haversine metric to a nationwide multi-province dataset. This project addresses that gap by applying DBSCAN spatial clustering, KDE density analysis, and Moran's I autocorrelation across all seven provinces of Nepal using a 5,000-record dataset, providing the first province-wide unsupervised spatial hotspot analysis of Nepal road accidents.

3. MATHEMATICAL FORMULATION

3.1 Haversine Distance Formula

Since accident locations are stored as GPS coordinates (latitude φ, longitude λ in degrees), the great-circle distance between two points is computed using the Haversine formula:

$$a = sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + cos(\phi_1) \cdot cos(\phi_2) \cdot sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)$$

$$c = 2arcsin\left(\sqrt{a}\right)$$

$$d = R \cdot c \qquad where\ R = 6371\ km\ (Earth's\ mean\ radius)$$

Coordinates are converted from degrees to radians before being passed to DBSCAN:

$$\phi_{rad} = \phi_{deg} \times \frac{\pi}{180}, \qquad \lambda_{rad} = \lambda_{deg} \times \frac{\pi}{180}$$

The DBSCAN eps parameter is expressed in radians. An eps of 0.005 radians corresponds to:

$$d_{km} = 0.005 \times 6371 \approx 31.8\ km$$

3.2 DBSCAN Formulation

DBSCAN classifies each point p in dataset D as follows:

1. **ε-neighbourhood:** $N_\varepsilon(p) = \{q \in D \mid dist(p, q) \leq \varepsilon\}$
2. **Core point:** p is a core point if $|N_\varepsilon(p)| \geq MinPts$
3. **Border point:** p is reachable from a core point but $|N_\varepsilon(p)| < MinPts$
4. **Noise point:** p is neither a core point nor density-reachable from any core point (label $= -1$)

In this study: $\varepsilon = 0.005\ rad \approx 31.8\ km$, $MinPts = 100$.

3.3 Severity Score

A weighted severity index quantifies the human impact of each accident:

$$Severity_Score_{raw} = 3 \times Fatalities + Injuries$$

Fatalities are weighted 3× relative to injuries, reflecting their irreversible nature. The raw score is then min-max normalized to the range [0, 1]:

$$Severity_Score = \frac{Severity_Score_{raw} - min}{max - min}$$

3.4 Risk Score

A composite risk score combines the normalized severity score with binary risk flags:

$$Risk_Score = w_1 \cdot Severity + w_2 \cdot High_Risk_Road + w_3 \cdot Bad_Weather + w_4 \cdot High_Risk_Cause$$

Where the binary flags are defined as:

$$High_Risk_Road = \{1 \ if \ Road_Type \in \{Hill\ Road, Ring\ Road\}\ 0 \ otherwise$$

$$Bad_Weather = \{1 \ if \ Weather \in \{Rainy, Foggy\}\ 0 \ otherwise$$

$$High_Risk_Cause = \{1 \ if \ Cause \in \{Drunk\ Driving, Overspeeding, Road\ Condition\}\ 0 \ otherwise$$

The Risk Score is also min-max normalized to [0, 1].

3.5 KDE — Kernel Density Estimation

KDE estimates the continuous probability density of accident locations. For n accident points $\{x_1, x_2, \dots, x_n\}$, the density at point x is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

Where K is the Gaussian kernel function and h is the bandwidth. In this study, bw_method = 0.05 is used via scipy.stats.gaussian_kde. A smaller h produces sharper, more localized hotspots; a larger h produces broader density regions.

3.5.1 Scott's Rule for Bandwidth Selection in KDE

In this study, the bandwidth parameter $h$for Kernel Density Estimation was determined using Scott's Rule, an automatic bandwidth selection method that minimizes the Mean Integrated Squared Error (MISE) between the estimated and true density functions. Scott's Rule is defined as:

$$h = n^{-1/5} \cdot \hat{\sigma}$$

where $n$is the number of observations and $\hat{\sigma}$is the standard deviation of the data. For two-dimensional spatial data, the rule generalizes to:

$$h_j = n^{-1/(d+4)} \cdot \hat{\sigma}_j$$

where $d = 2$ denotes the number of dimensions and $\hat{\sigma}_j$is the standard deviation along dimension j, yielding:

$$h_j = n^{-1/6} \cdot \hat{\sigma}_j$$

3.6 Moran's I — Spatial Autocorrelation

Moran's I tests whether similar accident severity values are spatially clustered or randomly distributed:

$$\boldsymbol{I} = \frac{\boldsymbol{n}}{\sum_i^n \sum_j^n \boldsymbol{w_{ij}}} \cdot \frac{\sum_i^n \sum_j^n \boldsymbol{w_{ij}}(\boldsymbol{x_i} - \boldsymbol{X})(\boldsymbol{x_j} - \boldsymbol{X})}{\sum_i^n (\boldsymbol{x_i} - \boldsymbol{X})^2}$$

Where n is the number of observations, x_i is the severity/risk score at location i, X is the mean score, and w_{ij} is the spatial weight between locations i and j.

Spatial Weights Matrix: A k-nearest neighbours (KNN) matrix with $k = 8$ is used, row-standardized so each row sums to 1:

$$w_{ij}^{std} = \frac{w_{ij}}{\sum_j^n w_{ij}}$$

Interpretation:

- $I > 0$ → Positive spatial autocorrelation (hotspots exist)
- $I \approx 0$ → Random spatial pattern
- $I < 0$ → Negative autocorrelation (dispersed pattern)

Statistical significance is assessed via permutation testing (999 simulations), with $p < 0.05$ as the threshold.

3.7 Silhouette Score — Cluster Validation

The Silhouette Score measures how well-separated the DBSCAN clusters are. For each clustered point i:

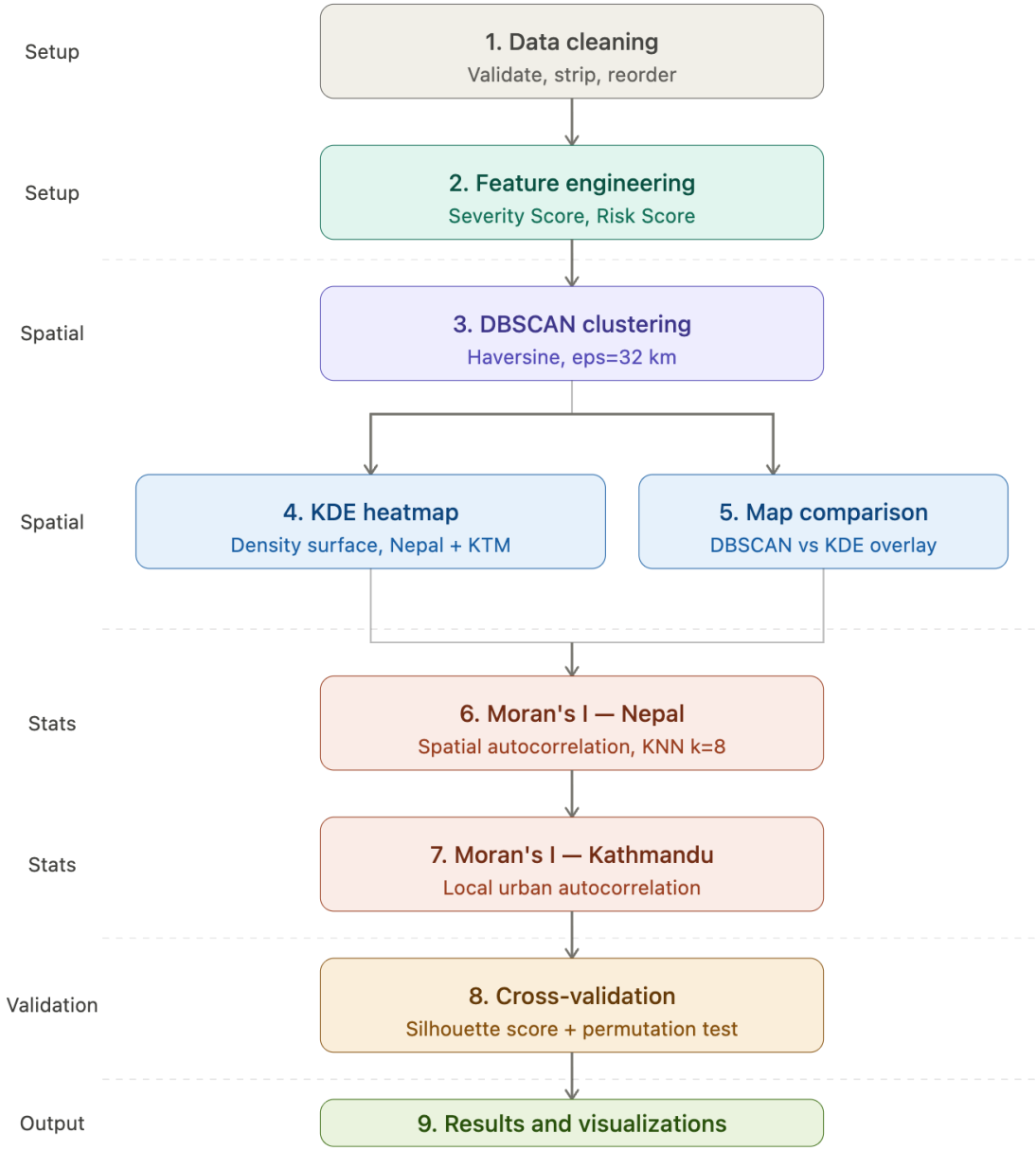
$$s(i) = \frac{b(i) - a(i)}{max\big(a(i),\ b(i)\big)}$$

Where $a(i)$ is the mean intra-cluster distance (cohesion) and $b(i)$ is the mean distance to the nearest other cluster (separation). The overall score is the mean of $s(i)$ across all non-noise points, computed using the Haversine metric.

- $s(i) \to 1.0$: Clusters are compact and well-separated
- $s(i) \to 0$: Clusters overlap
- $s(i) < 0$: Points may be misassigned

4. DATASET DESCRIPTION AND COLLECTION METHOD

4.1 Pipeline Overview



4.2 Data Collection

A large, structured, publicly accessible road accident dataset covering all seven provinces of Nepal does not currently exist in the format required for this analysis. Real accident data collected by Nepal Traffic Police is fragmented, inconsistently formatted, and unavailable at the granularity needed for spatial clustering and pattern analysis. As a result, the dataset used in this project was synthetically generated to reflect realistic accident distributions, modeled

after published Nepal Traffic Police annual statistics, provincial geographic characteristics, and road safety patterns documented in existing literature. The synthetic data was designed specifically to support DBSCAN spatial clustering, temporal analysis, and feature engineering.

4.3 Dataset Overview

- **Total records:** 5,000 accident records
- **Time period:** 2021 to 2025 (5 years)
- **Coverage:** All 7 provinces of Nepal across 30+ districts
- **Format:** Structured CSV with one row per accident record

4.4 Attributes

Each record contains the following fields:

- **Accident_ID** (String) — Unique identifier for each accident record
- **Year / Month** (Integer) — Year and month the accident occurred
- **Province / District** (String) — Geographic province and district of the accident
- **Latitude / Longitude** (Float) — GPS coordinates in decimal degrees, validated within Nepal's bounds (Lat 26°–30°N, Lon 80°–88°E)
- **Road_Type** (String) — Type of road: Urban Road, Hill Road, Highway, Rural Road, or Ring Road
- **Vehicle_Type** (String) — Primary vehicle involved: Motorcycle, Car, Bus, Truck, Tempo, or Bicycle
- **Accident_Cause** (String) — Primary cause: Human Error, Overspeeding, Drunk Driving, Weather, Flood, Landslide, or Road Condition
- **Time_of_Day** (String) — Time period: Morning, Afternoon, Evening, or Night
- **Weather_Condition** (String) — Weather at time of accident: Clear, Rainy, or Foggy
- **Injuries** (Integer) — Number of persons injured
- **Fatalities** (Integer) — Number of deaths
- **Emergency_Response_Time_Minutes** (Integer) — Minutes until emergency services arrived
- **Victim_Age_Group** (String) — Age bracket of the primary victim (Child, Young Adult, Adult, Middle Aged, Senior)
- **Victim_Gender** (String) — Gender of the primary victim

5. DATA CLEANING AND EXPLORATORY DATA ANALYSIS

5.1 Data Cleaning

Data cleaning was performed using Python (pandas) in the following steps:

- **Missing values and duplicates:** Zero missing values and zero duplicate rows confirmed.
- **Whitespace stripping:** All string columns stripped of leading/trailing spaces.
- **Coordinate validation:** All GPS coordinates verified within Nepal's bounds (Lat 26°–30°N, Lon 80°–88.15°E).
- **Numeric validation:** Injuries, Fatalities, Number of Vehicles, and Response Time confirmed to have no negative values.
- **Categorical validation:** All categorical columns validated against expected value sets to detect typos or unexpected categories.
- **Age group standardization:** Victim_Age_Group mapped to readable labels (e.g. 'Child (0-14)').
- **Date reconstruction:** Accident_Date synthesized from Year and Month fields. Month_Name column added for readability.
- **Column cleanup:** Dataset sorted chronologically, reordered as ID → Time → Location → Spatial → Incident → Victim, and SDG_Indicator dropped.

Result: 5,000 clean records with 23 columns, zero missing values, zero duplicates.

5.2 Exploratory Data Analysis

EDA was conducted using Matplotlib and Seaborn to understand the structure and distribution of the dataset before applying spatial methods. Analyses covered geographic distribution, accident causes, vehicle types, temporal patterns, severity and risk score distributions, feature correlations, victim demographics, and weather effects. Detailed findings from each analysis are presented in Results Interpretation.

6. FEATURE ENGINEERING

Two key engineered features were created to support spatial autocorrelation testing, cluster profiling, and correlation analysis:

- **Severity_Score:** Computed as (Fatalities × 3) + Injuries, then min-max normalized to [0, 1]. Fatalities are weighted three times more than injuries to reflect their irreversible nature.
- **Risk_Score:** A composite score combining Severity_Score with three binary risk flags — High_Risk_Road (Hill Road or Ring Road), Bad_Weather (Rainy or Foggy), and High_Risk_Cause (Drunk Driving, Overspeeding, or Road Condition) — then normalized to [0, 1].

7. SPATIAL ANALYSIS METHODS

7.1 DBSCAN — Spatial Hotspot Detection

- DBSCAN was applied to identify geographic accident hotspots using the Haversine metric for real-world distance computation.
- Coordinates were converted from degrees to radians before fitting.
- Parameters: eps = 32 km (0.005024 rad), min_samples = 100, algorithm = ball_tree, metric = haversine.
- Points not belonging to any cluster were labeled as noise (-1).
- The same procedure was repeated on the Kathmandu Valley subset with tighter parameters (eps = 0.5 km, min_samples = 30) to capture chowk-level hotspots.

7.2 KDE — Kernel Density Estimation

- KDE was applied to generate a continuous accident density surface using scipy.stats.gaussian_kde with bw_method = 0.05.
- Two layers were produced — raw accident density and a Severity_Score weighted version.
- Visualized as an interactive heatmap using Folium's HeatMap plugin.
- The same was repeated for Kathmandu Valley to produce a finer-resolution local density map.

7.3 Moran's I — Spatial Autocorrelation

- Moran's I was computed to test whether high-severity and high-risk accidents are spatially clustered or randomly distributed.
- A k-nearest neighbours spatial weights matrix (k = 8) was built using libpysal.weights.KNN and row-standardized.
- Statistical significance assessed at $p < 0.05$.

- Analysis was performed on both the full Nepal dataset (5,000 records) and the Kathmandu subset (1,126 records) to compare national vs. urban spatial autocorrelation patterns.

8. Model Evaluation and Validation

8.1 Silhouette Score — DBSCAN Validation

The Silhouette Score was computed using the Haversine metric to assess cluster quality.

- **Overall Score: 0.7027 — Strong cluster separation**
- Most clusters scored above 0.70, with Cluster 0 (Kathmandu) and Cluster 6 (Jhapa) achieving the highest scores of 0.9032 and 0.9384 respectively. Clusters 2 (Rupandehi) and 4 (Parsa) scored moderately at 0.4157 and 0.4538, indicating slight geographic overlap in the dense Terai corridor.

This confirms that the chosen parameters (eps = 32 km, min_samples = 100) produced geographically meaningful clusters.

8.2 Permutation Test — Moran's I Validation

Moran's I significance was validated using 999 permutation simulations under the null hypothesis of random spatial distribution.

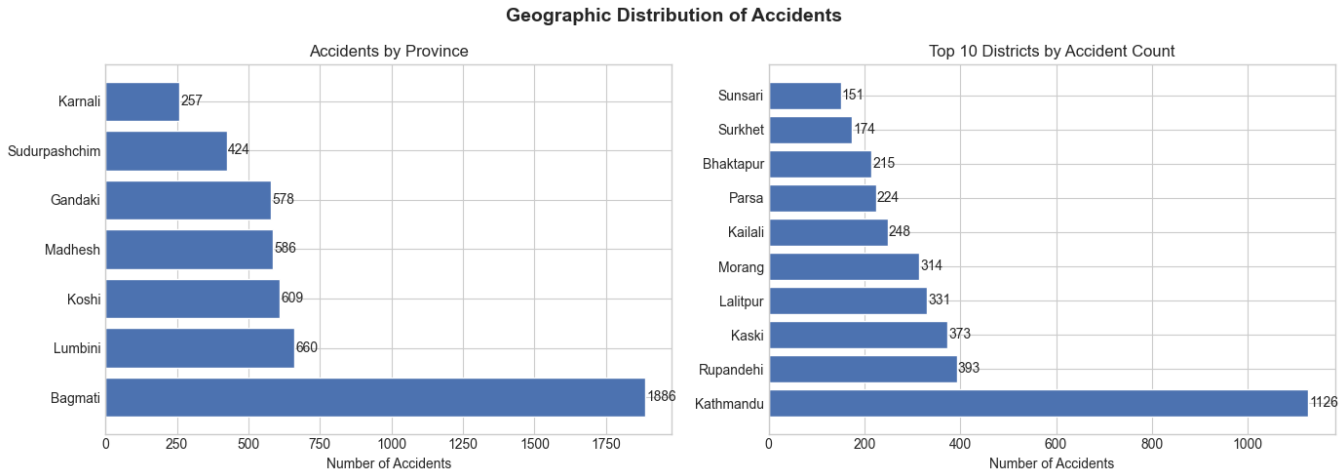
Variable	Moran's I	p-value	Result
Severity Score	0.0116	0.0430	✓ Significant
Risk Score	0.0652	0.0010	✓ Significant

Both scores show statistically significant positive spatial autocorrelation ($p < 0.05$), confirming that high-risk and high-severity accidents cluster geographically rather than occurring randomly. The null hypothesis of random distribution is rejected for both variables.

9. RESULTS AND INTERPRETATION

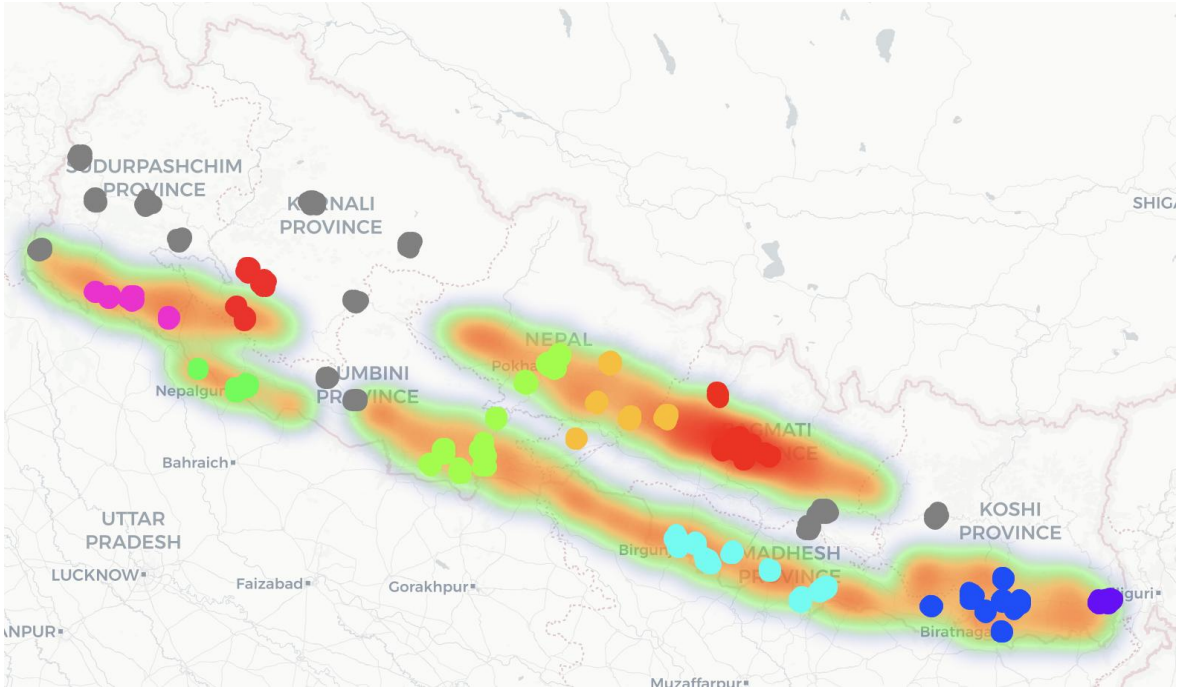
9.1 Geographic Distribution

Bagmati Province records the highest accident count (1,886), more than double any other province, driven by Kathmandu's high traffic density. At district level, Kathmandu (1,126) leads by a wide margin, followed by Rupandehi (393) and Kaski (373). Karnali (257) has the fewest accidents, reflecting its sparse population and low vehicle density.



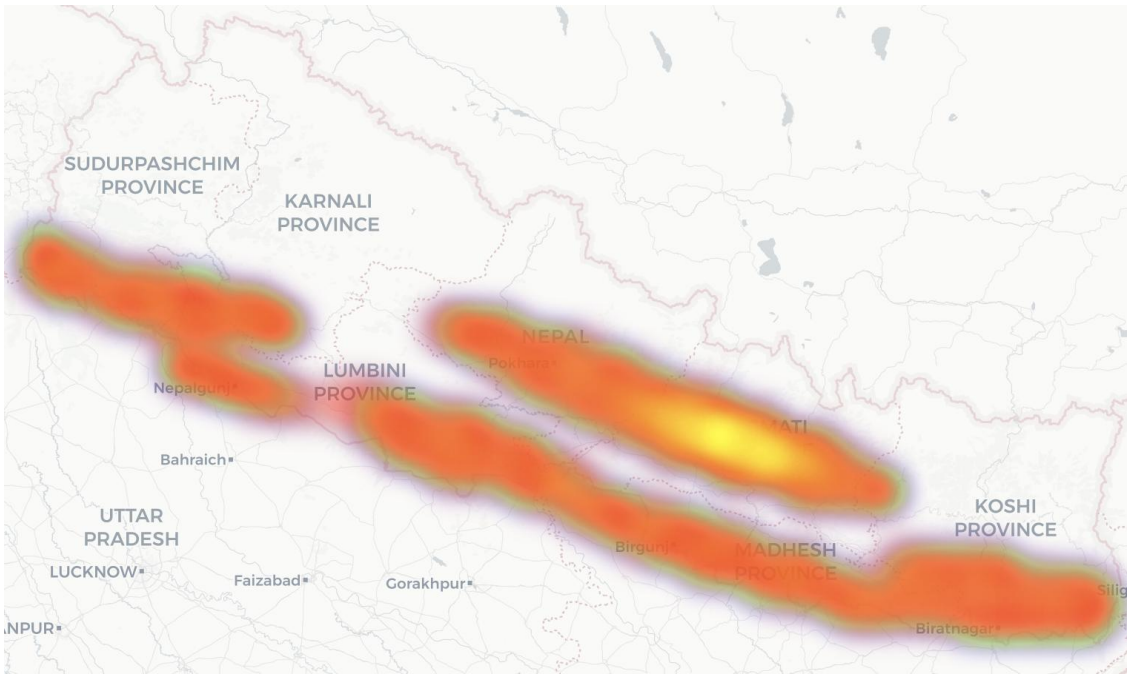
9.2 DBSCAN Spatial Clustering

DBSCAN identified 9 distinct accident clusters across Nepal with 374 noise points (7.5%). The largest cluster is centered on Kathmandu with 1,766 accidents. All clusters correspond to major urban centers or highway corridors. The combined DBSCAN + KDE map confirms that both methods agree on the same high-density regions across the Terai corridor and Kathmandu Valley.



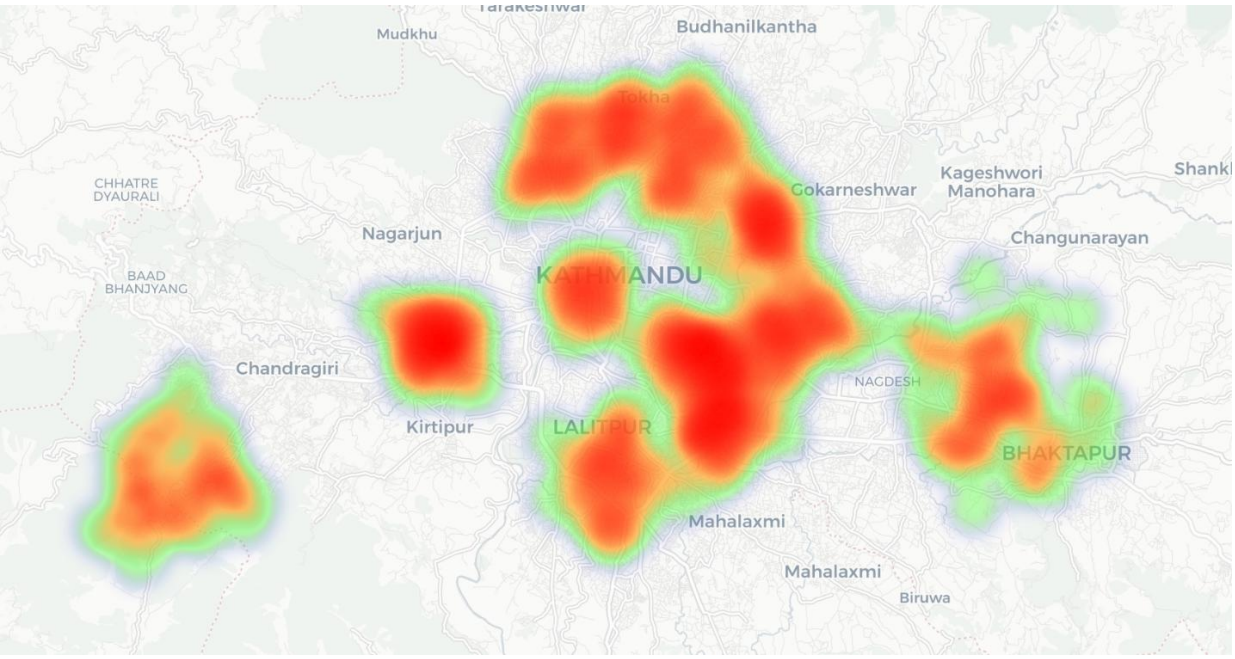
9.3 KDE Density Surface

The KDE heatmap reveals a continuous high-density band running across the southern Terai from west to east, with the most intense concentration centered around Kathmandu. The far-western Kailali/Dhangadhi region and eastern Morang/Biratnagar corridor appear as secondary hotspots.



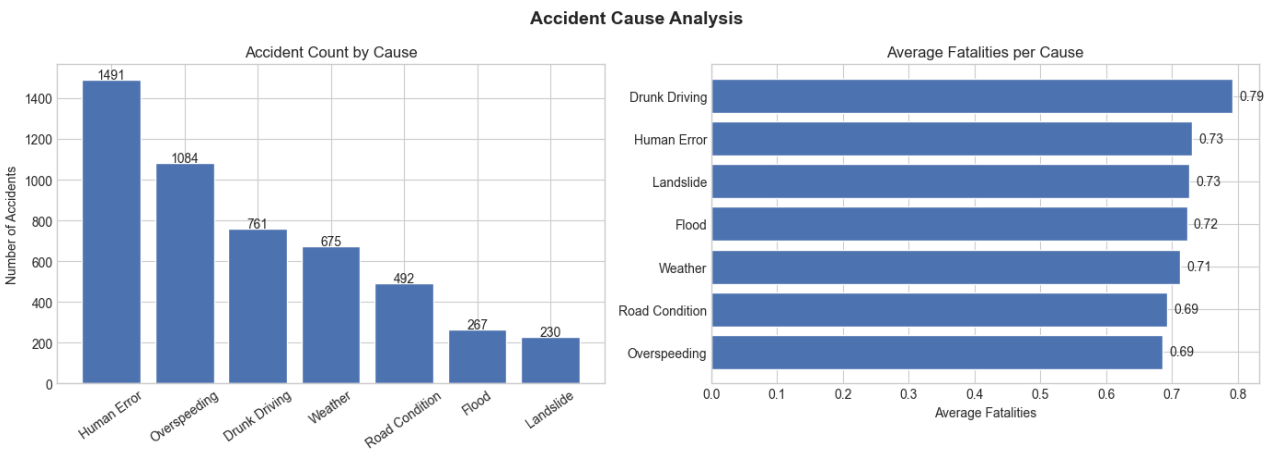
9.4 KDE — Kathmandu Valley

Within the Kathmandu Valley, the KDE map shows intense accident density along the Ring Road corridor and major chowks — particularly Kalanki, Koteshwor, and Lagankhel. The Kirtipur area appears as an isolated western hotspot. Peripheral residential areas such as Budhanilkantha and Mahalaxmi show notably lower density.



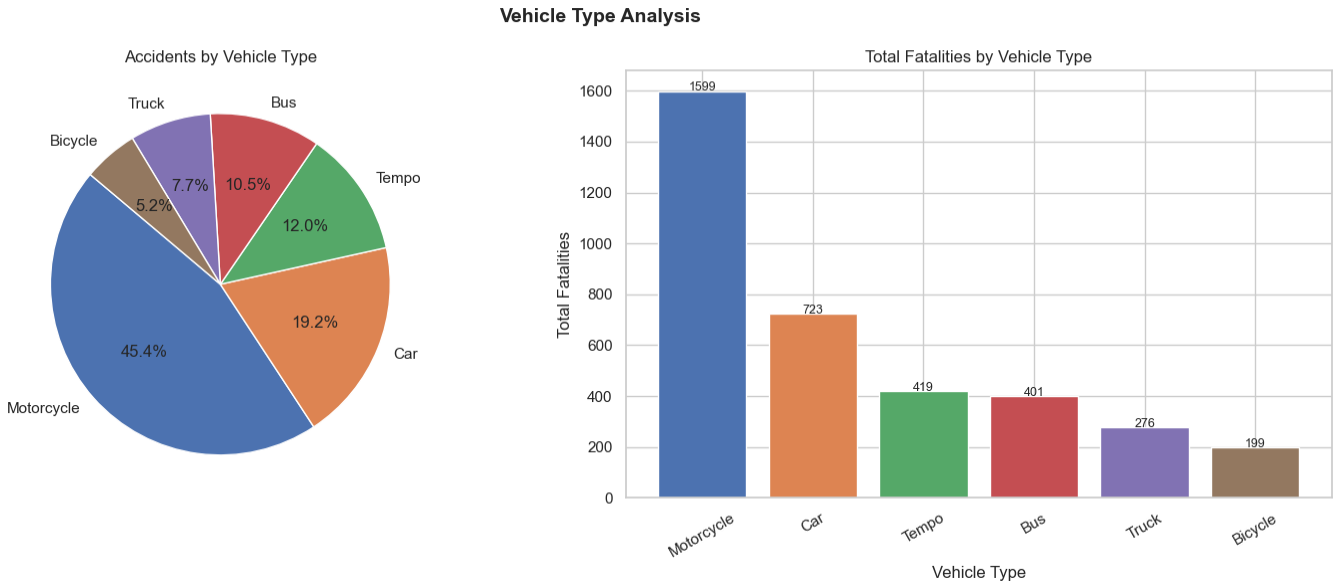
9.5 Accident Cause Analysis

Human Error leads with 1,491 accidents (29.8%), followed by Overspeeding (1,084) and Drunk Driving (761). However, Drunk Driving produces the highest average fatality rate (0.79) — meaning fewer incidents but more deaths per event. Behavioral causes combined account for 66.7% of all accidents nationally.



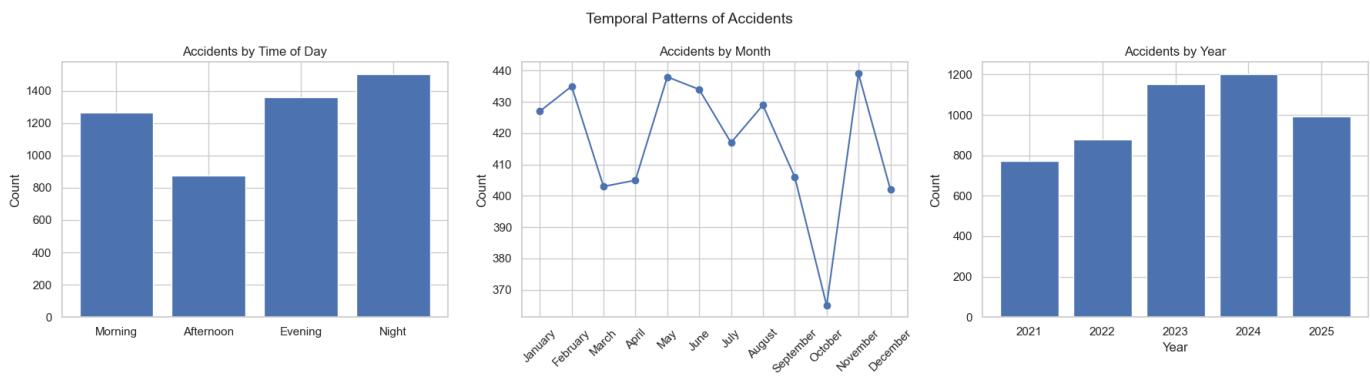
9.6 Vehicle Type Analysis

Motorcycles account for 45.4% of all accidents and the highest total fatalities (1,599) — nearly double that of cars (723). Buses and trucks produce significant fatalities relative to their accident share due to collision severity.



9.7 Temporal Patterns

Night (1,502) and Evening (1,360) are the highest-risk periods due to reduced visibility and lower enforcement. Monthly frequency is broadly stable (365–439), with a sharp October dip due to Dashain/Tihar holidays and a post-holiday spike in November (439). The yearly trend peaks in 2024 (1,201), up from 774 in 2021.

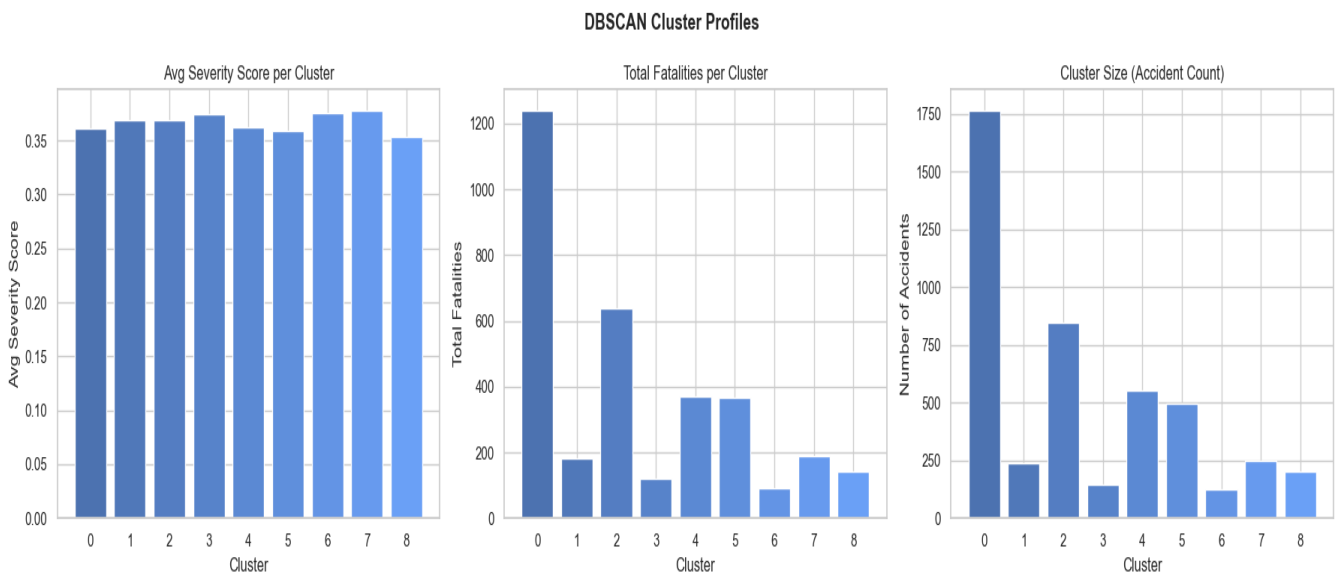


9.8 Severity and Risk Score Distributions

Severity scores concentrate between 0.2–0.6, indicating most accidents are moderate. Risk scores are right-skewed, peaking around 0.3–0.4. Severity scores are nearly identical across all road types (0.356–0.367), confirming road type alone does not determine accident severity.

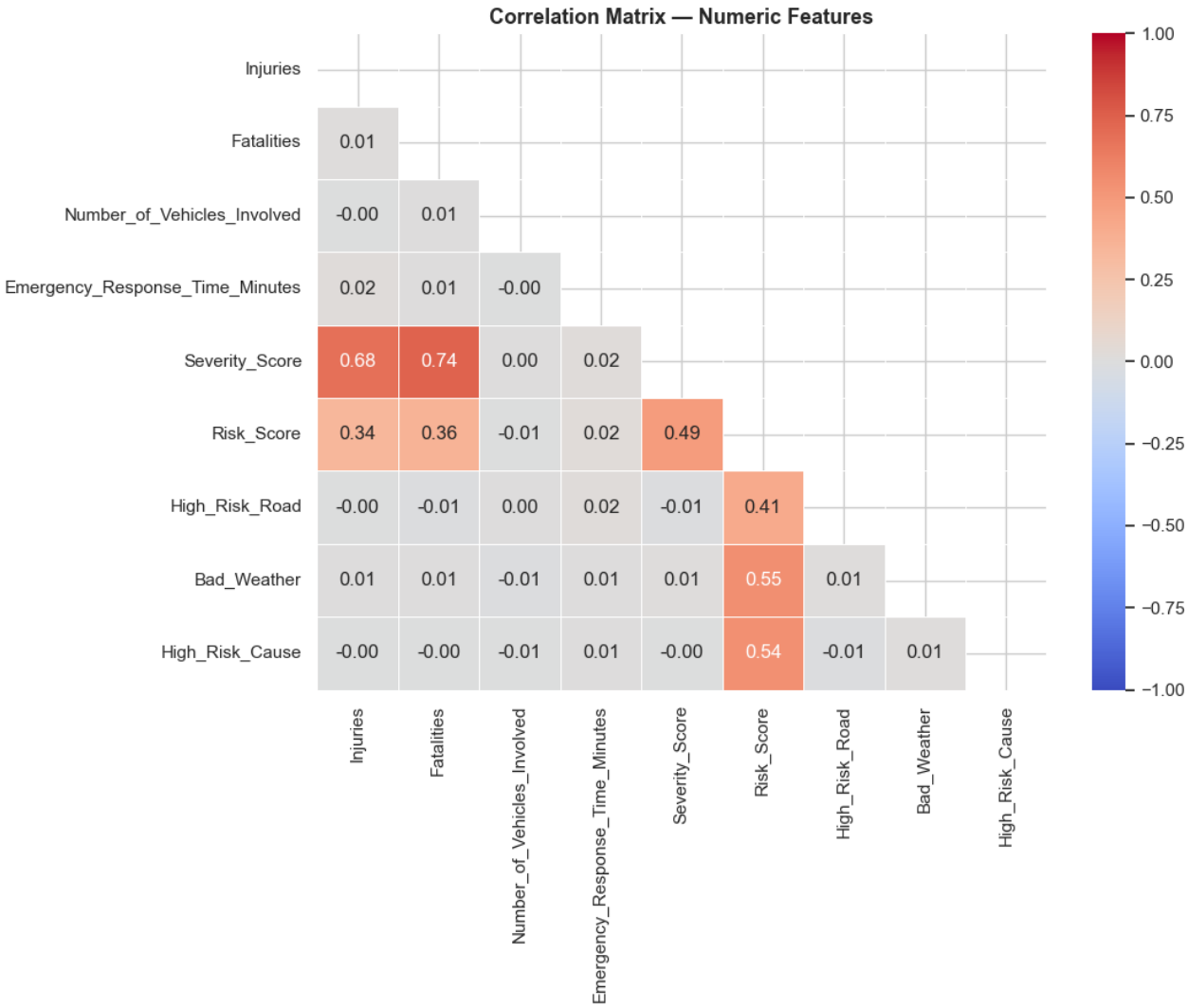
9.9 DBSCAN Cluster Profiles

Average severity is consistent across all 9 clusters (0.35–0.37), meaning no single cluster is disproportionately more severe per accident. Cluster 0 (Kathmandu) dominates in total fatalities (1,241) and accident count (1,766).



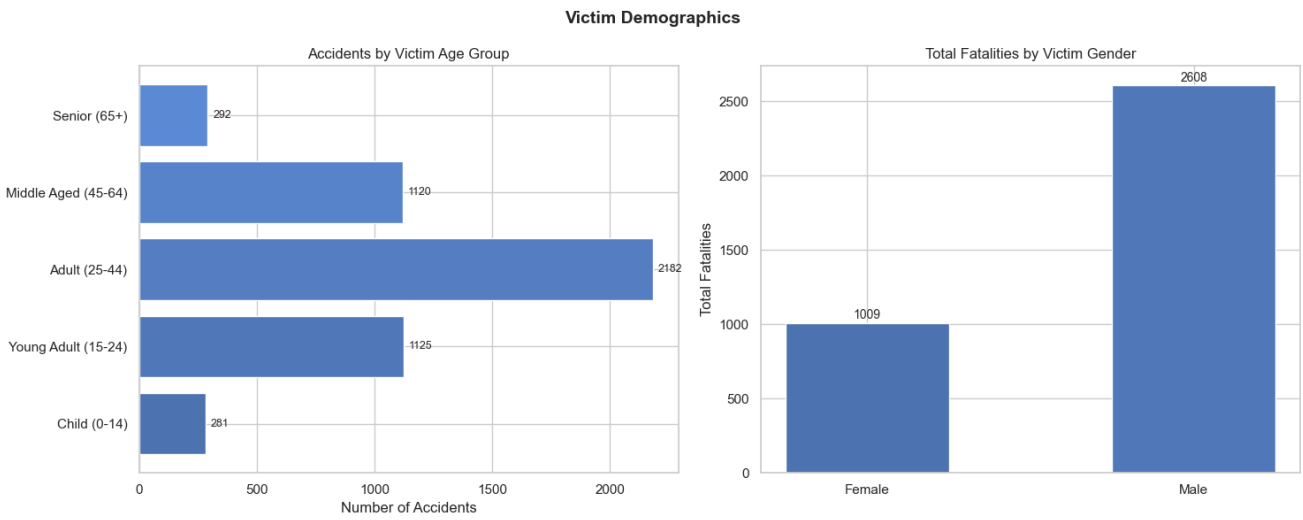
9.10 Correlation Analysis

Severity Score shows the strongest correlations with Fatalities (0.74) and Injuries (0.68). Risk Score correlates moderately with Bad Weather (0.55), High Risk Cause (0.54), and High Risk Road (0.41). Response time and number of vehicles show near-zero correlation with outcomes.



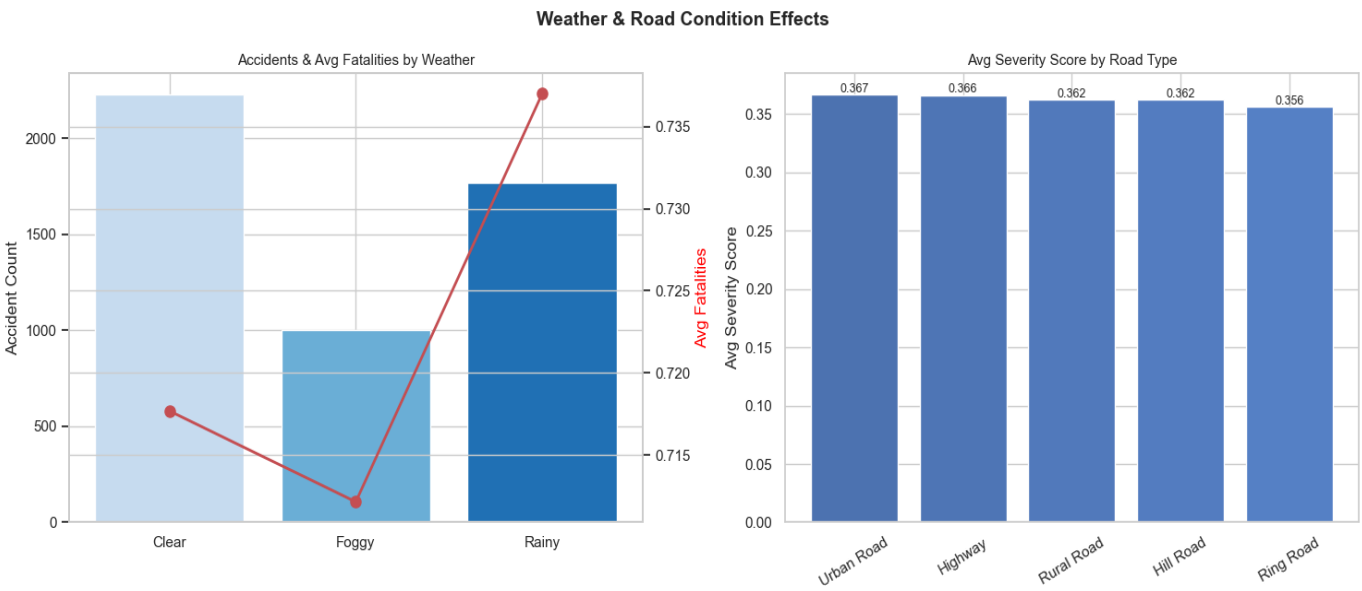
9.11 Victim Demographics

Adults aged 25–44 are the most affected group (2,182 accidents, 43.6%), followed by Young Adults 15–24 (1,125) and Middle Aged 45–64 (1,120). Male victims account for 2,608 fatalities vs 1,009 for females — a ratio of 2.6:1 — reflecting higher male exposure as motorcyclists and daily commuters.



9.12 Weather and Road Condition Effects

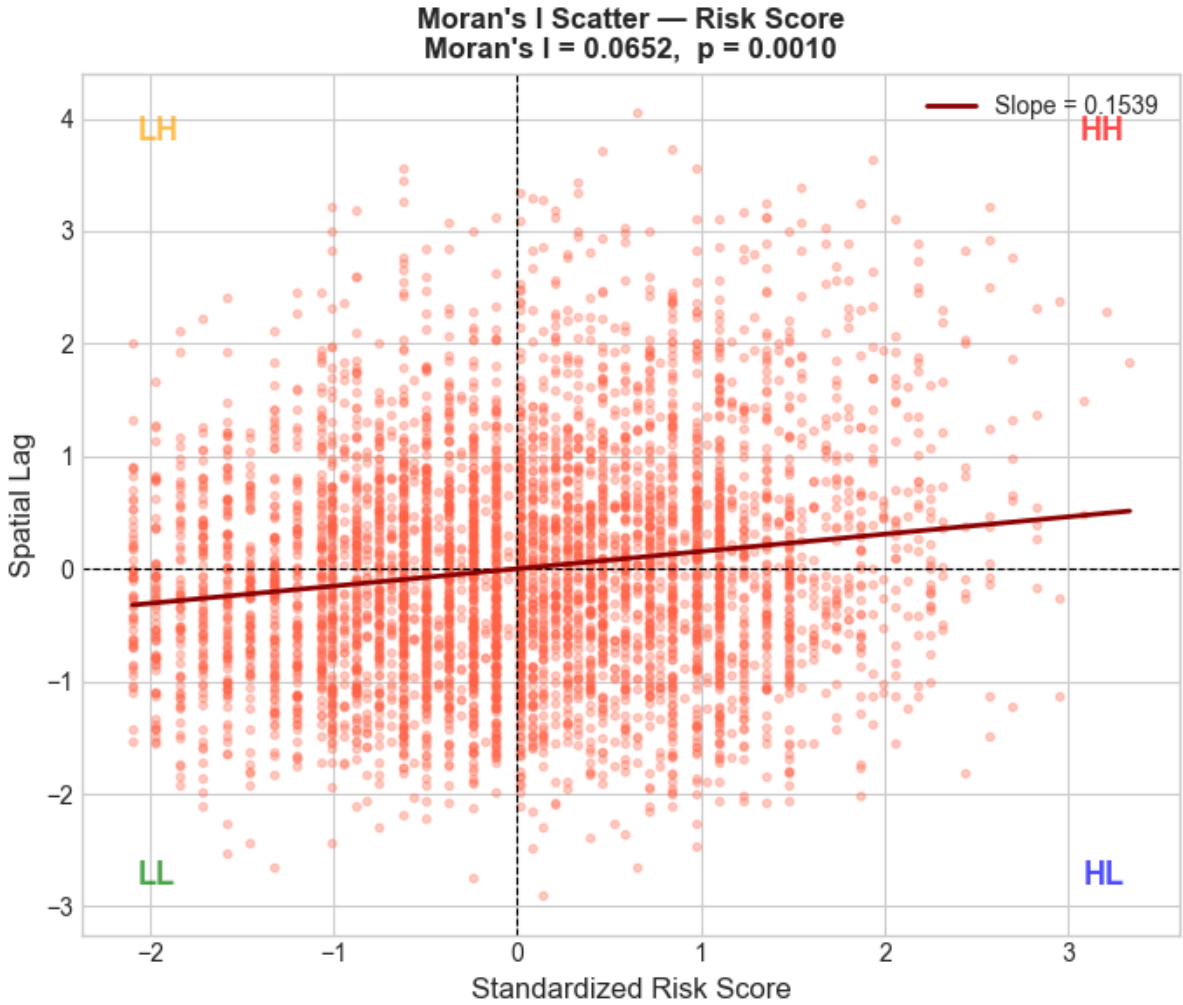
Clear weather records the most accidents (2,228) but rainy conditions produce the highest average fatality rate (0.74), confirming wet weather makes accidents deadlier. Severity scores are nearly uniform across road types, indicating road type has minimal independent effect on outcome.

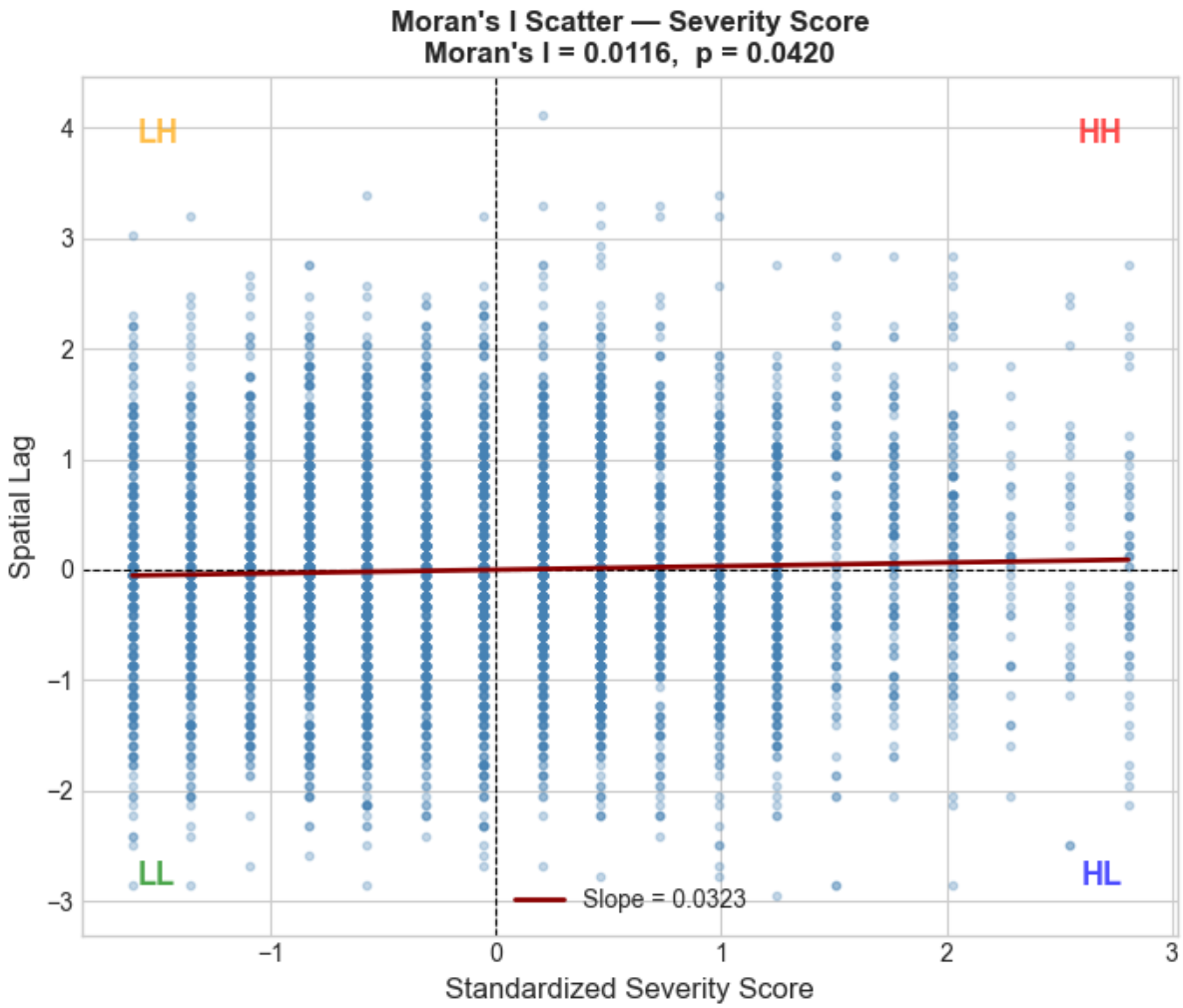


9.13 Moran's I — Spatial Autocorrelation

Both Severity Score ($I = 0.0116$, $p = 0.043$) and Risk Score ($I = 0.0652$, $p = 0.001$) show statistically significant positive spatial autocorrelation. The Risk Score scatter shows a more

pronounced HH (high–high) quadrant, confirming high-risk accidents genuinely cluster geographically. Severity is more weakly autocorrelated, suggesting severe accidents can occur anywhere but risk factors concentrate spatially.




Moran's I Scatter — Severity Score
Moran's I = 0.0116, p = 0.0420
LH
HH
LL
HL
Slope = 0.0323
Spatial Lag
Standardized Severity Score
4
3
2
1
0
−1
−2
−3
−1
0
1
2
3

10. CONCLUSION

This project presented a comprehensive spatial and temporal analysis of road traffic accidents across Nepal using a synthetically generated dataset of 5,000 records spanning 2021 to 2025. The study applied DBSCAN spatial clustering, Kernel Density Estimation, Moran's I spatial autocorrelation, and exploratory data analysis to identify accident hotspots, examine patterns, and derive evidence-based insights.

The key findings are summarized as follows:

- **DBSCAN identified 9 distinct geographic clusters** with a Silhouette Score of 0.70, confirming strong spatial separation. Kathmandu (Cluster 0) is the largest hotspot with 1,766 accidents and 1,241 total fatalities, followed by Rupandehi (848) and Parsa (554).
- **KDE confirmed the same hotspot regions**, revealing a continuous high-density corridor across the southern Terai and a concentrated urban hotspot within the Kathmandu Valley along the Ring Road and major chowks.
- **Behavioral causes dominate nationally** — Human Error (29.8%), Overspeeding (21.7%), and Drunk Driving (15.2%) collectively account for 66.7% of all accidents, making driver behavior the primary intervention target.
- **Motorcycles are the most dangerous vehicle type**, accounting for 45.4% of accidents and 1,599 fatalities — nearly double that of any other vehicle.
- **Night-time is the highest-risk period** (1,502 accidents, 30.0%), and accident frequency has grown steadily from 774 in 2021 to 1,201 in 2024, reflecting increasing vehicle volumes.
- **Moran's I confirmed statistically significant positive spatial autocorrelation** for both Risk Score ($I = 0.0652$, $p = 0.001$) and Severity Score ($I = 0.0116$, $p = 0.043$), validating that high-risk accidents cluster geographically rather than occurring randomly.

The findings collectively demonstrate that Nepal's road safety challenge is driven primarily by behavioral factors concentrated in high-density urban and highway corridors. Targeted interventions — including motorcycle safety enforcement, night-time road policing, drunk driving campaigns, and improved emergency response in Lumbini and Sudurpashchim — represent the highest-priority actions supported by this analysis.

A key limitation of this study is the use of synthetic data, which, while modeled on published Nepal Traffic Police statistics, cannot fully capture the complexity of real-world accident patterns. Future work should incorporate real accident records as they become available, extend the spatial analysis to all 77 districts, and apply predictive modelling to forecast accident hotspot evolution over time.

10.1 Recommendations

Province-Specific Interventions

Province (Cluster)	Primary Problem	Recommended Intervention
Lumbini - Rupandehi	Human error, highway speeding, night accidents	Speed enforcement on Siddhartha Hwy, driver training, night safety signage
Sudurpashchim - Kailali	Human error, night accidents, slow response (48.1 min)	Ambulance stations, road lighting on Mahendra Hwy, night patrols
Bagmati - Kathmandu	Human error, urban road condition, high volume	Road maintenance program, pothole repair, urban speed monitoring
Koshi - Morang	Human error, night accidents (46.5 min response)	Additional trauma centers, GPS-enabled emergency dispatch
Madhesh - Birgunj	Human error, low fatality rate, border traffic	Driver licensing reform, road safety campaigns at Birgunj border
Karnali - Surkhet	Human error, remoteness, slow response (46.1 min)	Helicopter emergency access, district trauma centers, road barriers
Gandaki - Kaski	Human error, overspeeding, hill roads	Speed cameras on Pokhara roads, hill road safety barriers

10.2 Future Work

Future research directions include:

- Gathering Real Dataset from Nepal Police and Department of Roads by submitting a Research Proposal
- Expanding the dataset to include all districts within each province for more granular district-level analysis.
- Applying Getis-Ord Gi* for comparison with DBSCAN results.
- Incorporating road network data to identify specific dangerous road segments rather than geographic point clusters.
- Seasonal disaggregation of cluster analysis to map how hotspot locations shift between monsoon and dry seasons.
- Machine learning severity prediction models using the engineered features to predict fatal vs. non-fatal accident outcomes.

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